Supplementary Material
Do VLMs See What Blind Users Show Them? A Distribution-Shift and Evaluation-Protocol Audit on VizWiz

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This supplementary material collects the reproducibility details for the main paper. The accompanying code, the exact prompts, the per-prediction model outputs, and the per-condition scoring tables are included in this supplementary archive and reproduce every number in the main paper from the raw predictions.

1 Reproducibility details

Prompt template. Every model received the identical user message below; only the question, and (except in the blind control) the image, were substituted.

> *{question}*
> *Answer in a few words. If the image does not allow answering, reply 'unanswerable'.*

Decoding is greedy (temperature 0, no sampling); the maximum new-token budget was 64 tokens.

Answer matching. Strict scoring is the standard VQA exact-match soft accuracy, $\min(m/3, 1)$ over the ten reference answers [1], under a normalisation that lower-cases, strips punctuation, removes the articles {a, an, the}, and maps the common contractions (*e.g.* `dont`$\rightarrow$ `don't`). *Lenient* scoring applies the same normalisation and counts a reference as matched if it is a whitespace-delimited token substring of the prediction, or if the prediction's leading token is `yes`/`no` and equals the reference, then takes the same $\min(m/3, 1)$. The lenient matcher is an upper bound and is reported only as the top of a bracket. *Token-F1* is the standard normalised token-level F1 (the same normalisation), taken as the maximum over the ten reference answers; it is a principled graded matcher used to test whether the model ranking is stable across reasonable matchers rather than only between strict and the lenient upper bound.

Prompt sensitivity. To check whether the abstention behaviour in Section 4.2 of the main paper is a property of the models or of the prompt, we re-ran the clean VizWiz evaluation on a fixed 1500-question subset (the first 1500 of the validation order, identical across all variants) under four instructions: the default; the same with the abstention clause removed; a permissive abstention clause; and a strict one. We report the abstention rate, the answerable-subset accuracy, and the Kendall τ of the aggregate ranking against the default prompt.

Blind control. The blind condition removes the image and passes the question through each model's text-only chat template; the resulting answer is scored identically to a sighted one (same matcher and abstention detector). A model that errors on the text-only path yields a dropped item rather than a forced wrong answer, so the blind floor is not inflated by input-format failures.

Image versus question decomposition. To split the curated-to-real gap we ran both datasets in clean and blind conditions under a single no-abstention prompt ("Answer in a few words.") on a fixed 1500-item subset, so a blind score reads the language-prior floor rather than an abstention rate. The gap decomposes exactly as gap $= [\mathrm{VQAv2}_b - \mathrm{VizWiz}_b] + [(\mathrm{VQAv2}_s - \mathrm{VQAv2}_b) - (\mathrm{VizWiz}_s - \mathrm{VizWiz}_b)]$; confidence intervals are a 10^4-sample bootstrap resampling the nine models. Four models answer imageless VQAv2 from the prior (VQAv2-blind 0.38–0.45); the rest still decline it even without an abstention clause, which floors their VQAv2-blind score, so the all-model question axis is a lower bound.

Verbose-answer penalty (Gemma-3-12B). On identical VQAv2 questions, Gemma-3-12B returns a correct but complete-sentence answer that strict exact-match scores as zero, where a terse model scores one. Three examples (same question, both predictions verbatim) are shown below; in each the ground-truth answer is recoverable as a substring of the Gemma prediction, which is why its accuracy more than doubles under the lenient matcher.

Gemma-3-12B (strict 0)	LLaVA-1.6-7B (strict 1)	ground truth
"Yes, it is sweet."	"Yes"	yes
"No, a jacket."	"No"	no
"No, zebras are fighting."	"No"	no

Corruption subset. The corruption sweep is applied to a fixed subset of VizWiz-val: the deterministic first 1500 questions in validation order (seed-governed and identical across all models and severities), of which 886 are answerable (a 59% answerable share in this slice; the 32.1% unanswerable figure quoted in the main paper is for the full validation split, and answerability is not uniform in validation order). The degradation analysis is reported on those 886 answerable questions, where degradation should hurt. The subset is reproducible from the released configuration (condition `subset_items: 1500`, `seed: 0`).

2 Additional analyses

VizWiz answer types versus the answerability split. Table 1 gives clean-condition accuracy on the answerable subset broken down by the benchmark's own answer types, alongside the abstention rate. The two axes measure different things: the answer-type axis separates easy from hard answer *formats*, while the abstention rate varies by a factor of three across models at comparable answer-type accuracy.

Table 1: Clean VizWiz accuracy on the answerable subset by the benchmark's own answer types, with the overall answerable-subset accuracy and abstention rate.

Model	yes/no	number	other	answerable	abstain
Qwen3-VL-8B	0.800	0.653	0.595	0.609	41.8%
Qwen2.5-VL-7B	0.732	0.569	0.570	0.581	42.0%
LLaVA-1.6-13B	0.826	0.569	0.515	0.536	29.2%
LLaVA-1.6-7B	0.826	0.493	0.500	0.522	27.4%
InternVL3-8B	0.634	0.493	0.470	0.481	33.9%
Pixtral-12B	0.812	0.472	0.448	0.473	27.5%
Idefics3-8B	0.802	0.382	0.447	0.469	12.9%
InternVL3.5-8B	0.680	0.625	0.380	0.404	24.2%
Gemma-3-12B	0.207	0.319	0.223	0.224	17.4%

Answerable-subset accuracy under each prompt. Table 2 reports answerable-subset accuracy and abstention rate for all four prompts of the sweep on the fixed 1500-question subset. P0 is the default, P1 removes the abstention clause, P2 is permissive and P3 strict. The level moves (means 0.461, 0.381, 0.476, 0.489); the ordering is more stable, with Kendall τ against P0 of 0.78 (P1), 0.83 (P2) and 0.89 (P3).

Coverage and risk. Reading each prompt as an operating point of a selective predictor, Table 3 gives coverage (the share of answerable questions the model attempts) and risk (the error rate on those attempts). The trade-off is real and prompt-controllable: the strict prompt lowers risk and coverage together, the no-abstention prompt drives coverage to one and raises risk. A full risk-coverage curve would need a graded confidence score, which the released per-prediction outputs do not carry; these are genuine operating points rather than a fitted curve.

What the aggregate ranking orders. Against the aggregate VizWiz ordering, Kendall τ is 0.89 for the answerable-subset ordering, 0.83 for accuracy on the unanswerable class and 0.78 for the abstention rate. Four of the nine models change rank between the aggregate and the answerable-only ordering: Idefics3-8B $8 \rightarrow 7$, InternVL3.5-8B $7 \rightarrow 8$, InternVL3-8B $4 \rightarrow 5$ and LLaVA-1.6-7B $5 \rightarrow 4$.

Table 2: Answerable-subset accuracy (acc) and abstention rate (abs) under the four prompts. Removing the abstention clause (P1) forces attempts on questions the model would otherwise decline, and accuracy on the answerable subset falls.

	P0 default		P1 none		P2 permissive		P3 strict	
Model	acc	abs	acc	abs	acc	abs	acc	abs
Qwen3-VL-8B	0.576	51.1%	0.467	3.8%	0.582	52.3%	0.599	57.1%
Qwen2.5-VL-7B	0.547	48.6%	0.370	0.7%	0.596	53.7%	0.579	56.7%
LLaVA-1.6-13B	0.552	40.0%	0.447	0.2%	0.586	57.2%	0.584	56.8%
LLaVA-1.6-7B	0.539	37.2%	0.454	0.5%	0.547	39.7%	0.559	44.1%
Pixtral-12B	0.481	34.3%	0.438	10.8%	0.483	36.5%	0.501	27.1%
Idefics3-8B	0.438	18.5%	0.424	0.2%	0.456	20.0%	0.436	12.7%
InternVL3-8B	0.423	37.5%	0.368	0.5%	0.433	39.9%	0.459	41.7%
InternVL3.5-8B	0.393	31.1%	0.348	0.6%	0.398	30.2%	0.448	39.0%
Gemma-3-12B	0.201	22.6%	0.116	0.0%	0.201	22.8%	0.238	23.1%
mean	0.461		0.381		0.476		0.489	

Table 3: Coverage and risk on the answerable subset under each prompt.

	P0		P1		P2		P3	
Model	cov	risk	cov	risk	cov	risk	cov	risk
Qwen3-VL-8B	0.730	0.462	0.999	0.534	0.711	0.447	0.649	0.418
Qwen2.5-VL-7B	0.738	0.501	1.000	0.630	0.679	0.429	0.639	0.443
LLaVA-1.6-13B	0.800	0.468	1.000	0.553	0.626	0.417	0.614	0.404
LLaVA-1.6-7B	0.816	0.482	1.000	0.546	0.799	0.475	0.752	0.459
Pixtral-12B	0.813	0.552	0.955	0.573	0.804	0.553	0.889	0.527
Idefics3-8B	0.930	0.585	1.000	0.576	0.920	0.569	0.962	0.576
InternVL3-8B	0.858	0.621	0.999	0.632	0.839	0.615	0.815	0.593
InternVL3.5-8B	0.893	0.647	0.998	0.651	0.897	0.638	0.813	0.601
Gemma-3-12B	0.932	0.834	1.000	0.884	0.924	0.840	0.924	0.797

3 Additional experiments prepared for the camera-ready version

These runs were added in response to the reviews. All use the same fixed 1500-question subset and the same matcher as the main paper, and all numbers are on the *answerable* subset unless stated otherwise.

Image-provenance controls. Table 4 complements the blind control by keeping an image in the input while breaking its relationship to the question: a uniform grey image, a patch-shuffled image, and a *mismatched* image taken from a different item. Paired mean drops against clean are -0.102 (grey), -0.085 (shuffled) and -0.205 (mismatched). A plausible but wrong photograph is the most damaging of the three, and models detect an empty image far more reliably than a misleading one.

Table 4: Image-provenance controls: accuracy on the answerable subset, with the overall abstention rate under the grey and mismatched conditions.

Model	clean	grey	shuffled	mismatched	abst. grey	abst. mism.
Qwen3-VL-8B	0.609	0.394	0.468	0.309	97.9%	69.3%
Qwen2.5-VL-7B	0.581	0.384	0.441	0.289	97.1%	66.7%
LLaVA-1.6-13B	0.536	0.368	0.432	0.292	80.7%	54.8%
LLaVA-1.6-7B	0.522	0.358	0.466	0.296	74.7%	49.1%
InternVL3-8B	0.481	0.398	0.373	0.257	95.1%	61.0%
Pixtral-12B	0.473	0.332	0.379	0.258	82.4%	48.9%
Idefics3-8B	0.469	0.378	0.307	0.222	80.9%	35.4%
InternVL3.5-8B	0.404	0.390	0.344	0.229	90.7%	52.4%
Gemma-3-12B	0.224	0.227	0.171	0.153	60.7%	40.4%
mean	0.478	0.359	0.376	0.256		

Two further prompts. Table 5 separates the abstention instruction from the answer *format*. P4 keeps the abstention clause but asks for a word or short phrase; P5 keeps a free-form answer but asks the model to reserve "unanswerable" for genuinely uninformative images. P4 gives the highest mean of any prompt tested (0.508 against 0.461 for the default), and the gain concentrates on the one verbose model.

Table 5: Answerable-subset accuracy and abstention rate under the two additional prompts, with the default prompt (P0) repeated for reference.

	P0 default		P4 strict format		P5 calibrated	
Model	acc	abs	acc	abs	acc	abs
Qwen3-VL-8B	0.576	51.1%	0.602	50.9%	0.515	41.1%
Qwen2.5-VL-7B	0.547	48.6%	0.575	40.3%	0.515	42.9%
LLaVA-1.6-13B	0.552	40.0%	0.532	31.0%	0.494	32.0%
LLaVA-1.6-7B	0.539	37.2%	0.534	37.3%	0.478	16.1%
InternVL3-8B	0.423	37.5%	0.536	43.6%	0.388	30.7%
Pixtral-12B	0.481	34.3%	0.482	15.3%	0.496	40.1%
Idefics3-8B	0.438	18.5%	0.446	17.2%	0.434	13.8%
InternVL3.5-8B	0.393	31.1%	0.501	35.2%	0.291	22.7%
Gemma-3-12B	0.201	22.6%	0.363	11.5%	0.179	11.9%
mean	0.461		0.508		0.421	

Broader capture-failure suite. Table 6 reports six further corruption families at two severities (108 units). The negative result of the main paper survives: only severe misframing produces a clear loss. Every family raises the abstention rate, and because correct abstention is rewarded on the unanswerable class, 88 of the 108 corrupted cells have a *higher* aggregate VizWiz score than the clean run.

Table 6: Expanded corruption suite: paired mean change in answerable-subset accuracy, the worst-affected model, and the change in overall abstention rate.

Corruption	Sev.	Δ acc	worst model	Δ abstain
motion blur	mod.	+0.006	Pixtral-12B (−0.028)	+12.7
motion blur	sev.	+0.007	Pixtral-12B (−0.044)	+25.2
sensor noise	mod.	+0.005	Gemma-3-12B (−0.009)	+8.4
sensor noise	sev.	+0.015	Idefics3-8B (−0.011)	+13.2
glare	mod.	+0.003	Pixtral-12B (−0.012)	+8.7
glare	sev.	+0.003	Pixtral-12B (−0.026)	+10.3
occlusion	mod.	−0.003	Pixtral-12B (−0.028)	+9.1
occlusion	sev.	−0.020	Idefics3-8B (−0.052)	+14.2
low resolution	mod.	+0.010	Pixtral-12B (−0.020)	+10.8
low resolution	sev.	+0.003	Pixtral-12B (−0.034)	+21.4
misframing	mod.	−0.021	LLaVA-1.6-7B (−0.041)	+11.2
misframing	sev.	−0.067	Pixtral-12B (−0.114)	+21.5

Residual abstention credit on the answerable subset. The soft VQA metric credits "unanswerable" whenever any of the ten annotators gave that answer. Of the items labelled answerable, 46.0% carry at least one such annotation, worth 0.316 on average. Under a grey image, abstaining predictions on the answerable subset average 0.406 while attempted ones average 0.154. The answerability split therefore removes most of the abstention confound but not all of it; a stricter variant would drop every item with an unanswerable annotation, at the cost of 46% of the subset.

References

1. Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., Parikh, D.: Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In: CVPR. pp. 6904–6913 (2017)